HDOC- 11 – 1
NAME : DARSHIL RAJ
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Program to Find the Profit or Loss Percentage from Cost Price and Selling Price

This document provides a clean, well-commented implementation for finding the profit or loss percentage in C. The program takes the cost price and the selling price as input, decides whether there is a profit, a loss or neither, and calculates the percentage using an if–else if–else ladder.

1. C Implementation

Step 1: Understand the Logic

- If the selling price is greater than the cost price, there is a profit: Profit = Selling Price - Cost Price.
- If the selling price is less than the cost price, there is a loss: Loss = Cost Price - Selling Price.
- If both prices are equal, there is no profit and no loss.
- Profit % = (Profit / Cost Price) x 100 and Loss % = (Loss / Cost Price) x 100.
- Both percentages are calculated on the cost price. The cost price must be greater than 0, otherwise the division is not possible.

Step 2: Define Variables and Data Types

- Use float variables for the cost price, selling price, profit, loss and percentage, so decimal prices are accepted.

Step 3: Take User Input

- Display prompts using printf("Enter cost price: ") and printf("Enter selling price: "), then read the values using scanf("%f", &costPrice) and scanf("%f", &sellingPrice).

Step 4: Validate the Prices

- If the cost price is less than or equal to 0, or the selling price is less than 0, print an error message and do not calculate.

Step 5: Calculate and Display the Result

1. If sellingPrice > costPrice, calculate and print the profit and the profit percentage.
2. Else if sellingPrice < costPrice, calculate and print the loss and the loss percentage.
3. Otherwise (else), print that there is no profit and no loss.

Complete C Code

```c
/*
 * Q17 (User Inputs, Operations & Output)
 * Write a program to find profit or loss percentage given
 * the cost price and the selling price.
 */
#include <stdio.h>

int main() {
    float costPrice, sellingPrice, profit, loss, percentage;

    // Step 1: Read the cost price and selling price
    printf("Enter cost price: ");
    scanf("%f", &costPrice);
    printf("Enter selling price: ");
    scanf("%f", &sellingPrice);

    // Step 2: Validate the prices
    if (costPrice <= 0 || sellingPrice < 0) {
        printf("Invalid input! Enter valid prices.\n");
    }
    // Step 3: Profit, loss or neither
    else if (sellingPrice > costPrice) {
        profit = sellingPrice - costPrice;
        percentage = (profit / costPrice) * 100;
        printf("Profit = %.2f\n", profit);
        printf("Profit percentage = %.2f%%\n", percentage);
    } else if (sellingPrice < costPrice) {
        loss = costPrice - sellingPrice;
        percentage = (loss / costPrice) * 100;
        printf("Loss = %.2f\n", loss);
        printf("Loss percentage = %.2f%%\n", percentage);
    } else {
        printf("No profit, no loss.\n");
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc11_1.c -o hdoc11_1
$ ./hdoc11_1
Enter cost price: 500
Enter selling price: 600
Profit = 100.00
Profit percentage = 20.00%
```

Run 2:

```
$ ./hdoc11_1
Enter cost price: 800
Enter selling price: 600
Loss = 200.00
Loss percentage = 25.00%
```

Run 3:

```
$ ./hdoc11_1
Enter cost price: 250
Enter selling price: 250
No profit, no loss.
```

Run 4:

```
$ ./hdoc11_1
Enter cost price: 0
Enter selling price: 100
Invalid input! Enter valid prices.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: Cost Price = 500, Selling Price = 600
- Calculation:
  - Validation: 500 <= 0 or 600 < 0 is FALSE, so the input is valid
  - 600 > 500 is TRUE, so there is a profit
  - Profit = 600 - 500 = 100
  - Profit % = (100 / 500) x 100 = 0.2 x 100 = 20.00
- Output: Profit = 100.00, Profit percentage = 20.00%

Sample Test Case 2:

- Input: Cost Price = 800, Selling Price = 600
- Calculation:
  - Validation: 800 <= 0 or 600 < 0 is FALSE, so the input is valid
  - 600 > 800 is FALSE, and 600 < 800 is TRUE, so there is a loss
  - Loss = 800 - 600 = 200
  - Loss % = (200 / 800) x 100 = 0.25 x 100 = 25.00
- Output: Loss = 200.00, Loss percentage = 25.00%

Sample Test Case 3:

- Input: Cost Price = 250, Selling Price = 250
- Calculation:
  - Validation: the input is valid
  - 250 > 250 is FALSE and 250 < 250 is FALSE
  - The else block runs, so there is no profit and no loss
- Output: No profit, no loss.

Sample Test Case 4:

- Input: Cost Price = 0, Selling Price = 100
- Calculation:
  - Validation: 0 <= 0 is TRUE, so the input is invalid
  - The error message is printed and no calculation is done (this also avoids dividing by 0)
- Output: Invalid input! Enter valid prices.